IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2012

EEE PART I: MEng, BEng and ACGI

ENERGY CONVERSION

Tuesday, 29 May 10:00 am

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer All questions.
Q1 carries 40% of the marks. Questions 2 and 3 carry equal marks (30% each).

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : W.T. Pike
Second Marker(s) : B.C. Pal

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Special instructions for students

Physical Constants

permittivity of free space:	$\varepsilon_o = 8.85 \times 10^{-12}$ F/m
permeability of free space:	$\mu_o = 4\pi \times 10^{-7}$ H/m
charge of an electron:	$e = 1.6 \times 10^{-19}$ C

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The Questions

1.

a) An object is fabricated from a superconducting material containing an internal void. Can a surface charge exist on the internal surface? Justify your answers. [4]

b) Prove that the magnetic field within a solenoid, an electromagnet formed by winding a coil around a cylinder, is proportional to the number of coil turns per unit length of the coil. [4]

c) Why are x-rays capable of imaging the interior of the human body? [4]

d) A standard mains AC voltage of 240V is applied to an electrical load. The current is measured at 2A and leads in phase by 30 degrees. What is the real, reactive and apparent power consumed by the load? [4]

e) Considering their batteries as both a source and store of energy, how might electric vehicles be incorporated into a smart grid? [4]

f) An open circuit coil of 25 turns is coupled by a sinusoidally varying magnetic flux of magnitude 0.5 mWb. If a voltage of magnitude 5.0 V is generated at the coil terminals, what is the frequency of oscillation of the flux in rad/s? [4]

g) An inductor core has relative permeability $\mu_r = 800$, and contains a uniform magnetic field of magnitude $H = 2.0$ A/m. Calculate the energy density of the field, in J/m^3. [4]

h) Give two advantages of using a core of high relative permeability in an inductor. [4]

i) Calculate the inductance of a coil of 50 turns wrapped around a toroidal core of relative permeability $\mu_r = 1200$ with cross-sectional area 5.0 cm^2 and mean radius 6.0 cm. [4]

j) If a simple DC motor has no load torque at all, what will be its equilibrium speed ω for an applied armature voltage V_a? [4]

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2.

a) A 10 μF parallel-plate capacitor with a plate spacing of 0.1 mm of air is charged to a voltage of 12 V. Ignoring fringing fields, what is the force between the plates? Explain if the force would be larger or smaller if fringing fields were not ignored. [8]

If the gap is filled by a dielectric of relative permittivity 10, how does this force change? [2]

b) The dielectric is refilled with a lossy dielectric which has a relative permittivity $5 - 0.001j$ at a frequency of 10 MHz. Show that the capacitor acts like an ideal capacitor and a resistor in parallel for a 10 MHz applied AC voltage, and calculate their values. [10]

What proportion of the mean energy stored in the capacitor is expended in the resistor in one cycle of the voltage? [10]

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3. a) A transformer is represented by the equivalent circuit shown in Fig 3.1. The component values are:
$R_i = 2000\ \Omega$; $X_m = 4000\ \Omega$; $X_{t1} = 4.0\ \Omega$; and $R_{t1} = 5.0\ \Omega$;

the turns ratio N1:N2 = 4:1; and the load impedance is $Z_L = 6.0\ \Omega$. An input voltage V_1 of peak amplitude 100 V and frequency 100 Hz is applied.

Calculate the amplitudes and phases of I_1, I_2 and I_{in}. [6]

Calculate the average load power P_L and the efficiency η. [6]

b) For the circuit of Fig 3.1 with unknown component values, derive an algebraic expression for the value of the referred load Z_L', assuming the actual load is real ($Z_L = R_L$), in terms of R_i, X_m, X_{t1} and R_{t1}., that maximises the efficiency η. [6]

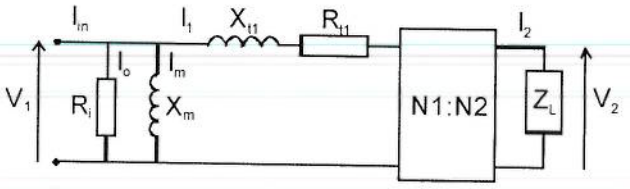


Figure 3.1

c) A simple DC motor has armature resistance $R_a = 10\ \Omega$, total flux $\Phi = 100$ mWb, $N = 20$ armature turns. If an input voltage $V_a = 30$V is applied, and the motor speed is 200 rpm, calculate:
the armature current I_a, the back-EMF e_a, the torque T, and the mechanical power P_m. [8]

d) The motor of part (c) is now connected to a load with torque-speed characteristic $T_L = 5\omega$ Nm·s, with ω the motor speed in radians/s. Find the equilibrium speed ω in this case if an input voltage $V_a = 60$ V is applied. [4]

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EE1-5

Answers 2 of 2 Energy Conversion 1/5

1 a) A perfect conductor will be at a single voltage. Hence, considering the definition of voltage, $V = \int \mathbf{E}.d\mathbf{l}$, there can be no field in the superconductor.

If there were both positive and negative charges on the surface of the void, this would imply that there were field lines between them, and hence the surface voltage would vary, which cannot be the case in a superconductor. However, it the charges were all the same polarity, there would have to be an overall flux into the void, but the void could not contain any charges of the opposing polarity to provide a destination for the flux. Hence no charging of the internal surface is possible. [4]

b) From Ampere's law $NI = \int \mathbf{H}.d\mathbf{l}$, where N is the number of turns of the coil, and I the current. Assuming the magnetic field is constant inside the coil and negligible outside, the integration becomes HL, where L is the length of the coil. Hence $H = IN/L$, and so is proportional to N/L, the number of turns per length of the coil. [4]

c) At high frequencies of an alternating electric field, even the lightest movable charges in the material, the electrons, cannot respond fast enough to the field. Hence the dielectric constant of the material, which reflects the interaction of the field with the material, approaches that of vacuum. [4]

d) Real power, P = IVcos30 = 416 W
Reactive power, Q = IV sin 30 = 240 VAr
Apparent power, S = IV = 480 W
[4]

e) Charge cars when generation capacity is available, and cheap, so mainly at night. This is consistent with usual travel patterns. During the day, when cars are not being used but are plugged in, they can act as a reservoir of energy by partially discharging their batteries back into the grid, though leaving enough power for expected travel later in the day. The use of cars for storage comes at a certain efficiency loss associated with the charge/discharge cycle. [4]

f) $V = Nd\Phi/dt = N\omega\Phi(t)$; $V_o = N\omega\Phi_o$; $\omega = V_o/N\Phi_o$; N = 25, $\Phi_o = 0.5\text{x}10^{-3}$, $V_o = 5$; $\underline{\omega = 40 \text{ rad/s}}$
[4]

g) $U' = \mu H^2 = \mu_r\mu_o H^2$; $U' = 800 \text{ x } 4\pi\text{x}10^{-3} \text{ x } 2^2$; $\underline{U' = 4.2\text{x}10^{-3} \text{ J/m}^3}$
[4]

h) i) The inductance is increased in proportion to μ_r, which can be very large, giving a much higher L for a given volume.
ii) The high permeability contains and guides the flux, which ensures that all turns link (nearly) all the flux, and reduces interference.
[4]

i) $L = N\Phi/I$; it's not necessary to derive the flux but if needed, use Ampere's law with a path around the torus giving $2\pi rH = NI$; $B = \mu H = \mu_r\mu_o NI/2\pi r$; $L = NBA/I =$

$\mu_r\mu_o N^2A/2\pi r$; Inserting the values, L = 1200 x 4πx10^{-3} x 2500 x 5 x$10^{-4}/2\pi$ x 6x x10^{-2} = 5 mH [4]

j) This can be done from memory or derived – with no load torque the armature current approaches zero, so $e_a = V_a = K\Phi\omega$ giving $\omega = V_a/K\Phi$ [4]

2. (a) Force, F, due to a charge Q in a field E is given by $F = QE$. The field between the plates of the parallel plate capacitor (away from the edge) will be $E = V/d$, where V is the voltage between the plates, and d is the spacing, while the charge is given by $Q = CV$, where C is the capacitance.

Hence $F = CV^2/d$ = 10 x 10^{-6} x 144/(0.1 x 10^{-3}) = 14.4 N [5]

If there is a fringing field, the lines will no longer be perpendicular to the plates, but will bulge outwards. Hence the field lines will be lengthened. Given the voltage between the plates is constant and given by $V = \int \mathbf{E}.d\mathbf{l}$, if the pathlength is longer, the field must be lower, and hence the force will be reduced. [3]

If the dielectric constant is ten times biiger, the capacitance will increase by the same amount, and from $F = CV^2/d$ so will the force. [2]

(b) $\varepsilon_r = \varepsilon_r' - j\varepsilon_r'' = 5 - j0.001$

$C = \frac{\varepsilon_0(\varepsilon_r' - j\varepsilon_r'')A}{d} = (\varepsilon_r' - j\varepsilon_r'')C_0$, where C_0 = 10 μF. [2]

The admittance Y is given by

$Y = j\omega C$

$= j\omega(\varepsilon_r' - j\varepsilon_r'')C_0$

$= j\omega\varepsilon_r' C_0 + \omega\varepsilon_r'' C_0$. [2]

Hence the impedance is of the form by

$$\frac{1}{Z} = \frac{1}{Z_c} + \frac{1}{R}$$

which represents an ideal capacitor and resistor in parallel where $Z_c = 1/j\omega\varepsilon_r' C_0$ and $R = 1/\omega\varepsilon_r'' C_0$. [3]

Therefore the capacitance of the ideal capacitor is given by $C' = \varepsilon_r' C_0$ = 50 μF [1]

and the resistance is given by $R = \frac{1}{\omega\varepsilon_r'' C_0}$ = 1/(2πx10) = 0.16 Ohms. [2]

Energy stored in capacitor is given by

$E_c = \frac{1}{2}C'V_{rms}^2$ [2]

Power expended in resistor is given by

$P_R = V_{rms}^2/R$

[2]

and hence the energy expended in one cycle of period T will be

$$E_R = \frac{TV_{rms}^2}{R} = \frac{2\pi V_{rms}^2}{\omega R}$$

[2]

The ratio is given by

$$\frac{E_R}{E_C} = \frac{2\pi V_{rms}^2}{\omega R \frac{1}{2} C' V_{rms}^2} = \frac{4\pi}{\omega R C'} = 4\pi \frac{\varepsilon_r''}{\varepsilon_r'} = 12 \times 10^{-3}$$

[4]

3. a)

$Z_L' = (N_1/N_2)^2 Z_L = 4^2 \cdot 6 = 96\ \Omega$

$I_{in} = I_1 + I_0 + I_m$

$I_1 = \dfrac{V_1}{96 + 5 + j4}$ $\quad |I_1| = \dfrac{100}{\sqrt{101^2 + 4^2}} = 0.989\text{A}$

$\measuredangle I_1 = -\tan^{-1}\dfrac{4}{101} = -2.3°$

$I_2 = (N_1/N_2) I_1 = 4 I_1 = 3.96\text{A} \angle -2.3°$

$I_0 = 100/2k = .05\text{A}$ $\quad I_m = 100/j4k = -j.025\text{A}$

$I_{in} = .989\cos 2.3° + .05 - j(.989 \sin 2.3° + .025)$

$= 1.040\text{A} \angle -3.53°$

$P_L = \frac{1}{2}|I_2|^2 Z_L = \frac{1}{2}(3.96)^2 \cdot 6 = 47.0\text{ W}$

$\eta = P_L/(P_L + P_{R_1} + P_c)$ $\quad P_c = \frac{1}{2}\frac{100^2}{2000} = 2.5\text{ W}$

$P_{R_1} = \frac{1}{2}|I_2|^2 R_1 = \frac{1}{2}(.989)^2 \times 5 = 2.45\text{ W}$

$\eta = \dfrac{47}{47 + 2.5 + 2.45} = 90.5\%$

b) In general, $\eta = \dfrac{P_L}{P_L + P_{R_1} + P_c}$ with $P_c = \dfrac{V_1^2}{R_c}$

(neglecting the common $\frac{1}{2}$) $\quad P_{R_1} = |I_1|^2 R_1$ $\quad P_L = |I_1|^2 R_L$

Let us normalize to $V_1 = 1$

$|I_1|^2 = \dfrac{1}{(R_1 + R_L')^2 + X_1^2}$ $\quad$ (note $Z_L' = R_L'$ here)

Then $\eta = \dfrac{R_L'}{R_L' + R_{R_1} + [(R_L + R_1)^2 + X_1^2]/R_c}$

3. b) (continued)

Multiply top and bottom by R_i', then take the derivative:

$$\frac{d\eta}{dR_L'} = \frac{R_i'\left(Q + (2R_1 + R_i')R_L' + R_L'^2\right) - R_L'R_L'(2R_1 + R_i' + 2R_L')}{(\text{denominator})^2}$$

Here we have introduced the constant $Q = R_1R_i' + R_1^2 + X_1^2$

Setting $d\eta/dR_L' = 0$: $\quad Q + (2R_1 + R_i')R_L' + R_L'^2 = R_L'(2R_1 + R_i' + 2R_L')$

Simplifying: $\quad Q = R_L'^2$

$$R_L' = \sqrt{R_1^2 + X_1^2 + R_1R_i'}$$

c) $e_i = K\Phi\omega$ (they should know this)

$K = 2N/\pi$

$= 2(20)/\pi = 12.7 \qquad \omega = \frac{200}{60} \times 2\pi = 20.93$

$e_i = (12.7)(0.1)(20.93)$

$e_i = \underline{26.67\ V} \qquad I_a = \frac{V_a - e_i}{R_a} = \frac{30 - 26.7}{10} = \underline{0.33A}$

$$T = \frac{P_m}{\omega} = \frac{e_i I_a}{\omega} = \frac{26.67 \times .33}{20.93} = \underline{0.424\ N\cdot m}$$

$P_m = e_a I_a = \underline{8.88\ W}$

d) $V_a = e_a + I_a R_a = K\Phi\omega + \frac{T}{K\Phi} R_a$

Here $T = 5\omega$ giving $V_a = \left(K\Phi + \frac{5R_a}{K\Phi}\right)\omega$

$K\Phi = 0.1K = 1.27$

$\omega = \frac{V_a}{\left(1.27 + \frac{5(10)}{1.27}\right)} = \frac{60}{40.7} = \underline{1.47\ rad/s}$